Omnipair Dusk: A Generalized Automated Market Maker for Oracle-less Spot, Credit, and Margin

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Abstract

Omnipair Dusk combines spot trading, collateralized credit, isolated margin positions, and funded liquidity vaults in one market on Solana. Its accounting identity separates cash, indexed cash-backed debt, and vault-owned reserve claims. Spot execution applies a second separation: unpaid cash-backed interest is excluded from curve principal, and hLP ownership is removed to obtain the ordinary liquidity tranche on which traders execute. A full-range constant-product tail and two nested liquidity bands define a continuous, piecewise analytic reserve path. After a trade, the program reconstructs hLP ownership and opposite-asset funding algebraically. Credit uses conservative internal price and depth observations, while realized yield, recenter funding, and loss recovery have distinct accounting paths. This paper describes the implemented mechanisms and their limitations, with reproducible numerical illustrations of the curve, interest controller, and hLP recovery incentives.

Implementation scope. Revised October 23, 2026 for the candidate integration of PR #52 at `2f46e0e7` with PR #45 at `04941f19`. Section 7 distinguishes implemented hLP accounting and recovery from remaining validation. Dusk is under active development and has not been deployed to production. Equations below suppress token normalization and integer rounding unless stated otherwise. The source map and generated examples accompany both drafts; numerical illustrations are not an audit, market backtest, or return forecast.

1 Introduction

The market functions are spot trading, collateralized credit, and isolated margin trading. A margin position combines a credit draw with a spot trade; hLP vaults separately provide funded exposure to liquidity shares. In the derivations, leverage denotes gross exposure divided by equity.

A spot venue, a credit market, and a liquidation route often hold separate views of the same asset. A lender may value collateral at a linear price even though recovery must occur through a finite pool. For a thinly traded pair, the difference between that mark and executable

recovery can dominate the economics of the loan. External price feeds can improve information, but do not create the inventory required to liquidate a position.

Dusk develops Omnipair's Generalized Automated Market Maker (GAMM) approach: spot and credit share a pair-local balance sheet [1]. Drawing credit replaces available cash with a debt claim, while trading changes the inventory coordinate. The market also maintains two aggregate hLP vaults, each targeting exposure to one asset through a funded position in the market's own liquidity shares. The intended benefit is a common accounting and settlement system for activities that otherwise depend on separate venues.

The current design has three important separations. First, stored reserve claims differ from cash available for a transfer. Second, the coordinate used to mint and burn liquidity shares differs from the ordinary tranche used to quote a spot trade. Third, collateral admission, liquidation eligibility, and actual liquidation proceeds use different valuations. These distinctions are necessary once an AMM contains debt, concentrated liquidity, and funded ownership at the same time.

"Oracle-less" means that the program does not require an external price feed to execute these mechanisms. It does use internal price observations, moving averages, depth measurements, and a persistent concentration center. Their quality depends on the market's trading history. Market-local accounting contains a loss within its accounting domain; it does not establish an external fair price or guarantee that liquidity remains available.

1.1 Related mechanisms

StableSwap demonstrated how an invariant can allocate more liquidity near an expected exchange rate [2]. CryptoSwap developed internal price adaptation and an economic budget for moving concentrated liquidity [3]. Dusk uses a different curve construction: a nonzero constant-product tail plus nested core and shoulder ranges, with an explicit custody bucket for retained recenter funding. The mathematical form and financing of that movement must therefore be evaluated on their own terms.

The adaptive rate-at-target idea is closely related to Morpho's AdaptiveCurveIRM [4]. Dusk implements its own bounded, discrete anchor update and accrues a shared asset-side debt index. Similar controller intuition does not imply identical integration, parameters, or numerical behavior.

Leveraged liquidity has a separate lineage. Egorov's treatment of constant-product LP payoff curvature motivates one-sided exposure through dynamically maintained leverage [5]. Yield Basis holds Curve LP collateral against crvUSD debt; its LEVAMM and VirtualPool support the rebalance mechanism [6]. Dusk represents the LP ownership and funding inside the same market and reconstructs them at the quoted endpoint. Section 5 makes the scope of that distinction precise.

1.2 What is established here

The reserve identities and endpoint formulas are accounting statements. Their continuous forms can be derived and checked independently. Whether the Rust program preserves them under every token decimal configuration, overflow boundary, account combination, and instruction sequence is an implementation question. Whether a configured market offers useful credit and sustainable LP returns is an empirical question. The examples at the end address selected arithmetic properties, while the expanded appendices connect those properties to source and describe the remaining evaluation work.

2 The GAMM Invariant

Consider base and quote assets, denoted b and q. For each side i, write C_i for accounted reserve cash, D_i^f for indexed public credit debt, D_i^ℓ for indexed isolated margin debt, and H_i for the aggregate hLP live-reserve component. The stored live reserve is

$$\boxed{R_i = C_i + D_i^f + D_i^\ell + H_i.} \tag{1}$$

Here $D_i^{\mathrm{cb}} = D_i^f + D_i^\ell$ is all cash-backed debt. Isolated margin is a separate health domain, but is already included in cash-backed debt. hLP funding debt D_i^h affects utilization and vault equity; it is not added again to the cash-backed term.

This is the GAMM reserve invariant from the earlier paper, with the debt domains made explicit. A reserve can be represented by available cash, a claim on a borrower, or an hLP live-reserve entry. Every successful transition must explain how those components account for the resulting live reserve. With no hLP, $H_i = 0$; with no debt either, the identity reduces to the familiar cash reserve of an unfunded AMM.

2.1 Connecting the notation

Earlier notation	Here	Meaning
$R_{\text{live,i}}$	R_i	Stored live reserve and proportional share basis.
$R_{\text{cash,i}}$	C_i	Accounted cash available subject to settlement checks.
$D_{\text{cash-debt,i}}$	$D_i^f + D_i^\ell$	Indexed public and isolated cash-backed debt.
$R_{\text{hLP,i}}$	H_i	hLP live-reserve contribution to the ledger.

Two further coordinates connect that ledger to a quote: X_i removes unpaid cash-backed interest, and (U, V) separates ordinary yLP inventory from hLP ownership. They refine the original invariant without replacing it. In particular, H_i is not the entire inventory owned by hLP: a vault's target-asset contribution also has cash or other reserve backing. Section 5 makes its target contribution and funded opposite claim explicit.

Let P_i^f and P_i^ℓ be the outstanding principal counters. Unpaid cash-backed interest is

$$J_i = \left[D_i^f - P_i^f\right]_+ + \left[D_i^\ell - P_i^\ell\right]_+ . \tag{2}$$

The executable reserve coordinate before separating hLP ownership is

$$X_i = R_i - J_i. \tag{3}$$

Thus interest accrual can increase the stored live reserve without expanding the curve principal. Using R_q/R_b as a universal spot price would miss both this adjustment and concentrated-curve geometry.

2.2 The invariant as a transition rule

Define the ledger residual $\mathcal{I}_i = R_i - C_i - D_i^{\mathrm{cb}} - H_i$. The invariant is $\mathcal{I}_i = 0$ before and after an admitted operation, equivalently

$$\boxed{\Delta R_i = \Delta C_i + \Delta D_i^{\mathrm{cb}} + \Delta H_i.} \tag{4}$$

The following table isolates elementary accounting steps in exact arithmetic. It holds hLP state fixed; a full spot trade or liquidity instruction can compose these steps with funding, ownership, fee, and controller transitions.

Accounting step	ΔC_i	ΔD_i^{cb}	ΔH_i	ΔR_i	ΔX_i
Principal credit draw a	$-a$	a	0	0	0
Principal repayment a	a	$-a$	0	0	0
Unpaid interest accrual j	0	j	0	j	0
Interest realization ι	0	$-\iota$	0	$-\iota$	0
Net spot input a	a	0	0	a	a
Net spot output b	$-b$	0	0	$-b$	$-b$

In the interest rows, ΔJ_i equals the change in indexed debt; realized interest is routed to its own custody and distribution path. In the spot rows, "net" denotes the curve principal after separating fee destinations. For hLP operations, equation (4) still holds, but ΔH_i and the cash effects must come from the admitted funding and ownership reconstruction rather than an assumed depth multiplier.

The ledger invariant and the curve invariant answer different questions. The former accounts for reserve claims through every operation. The latter defines the exchange path at a prepared geometry: constant product in the CPMM limit, or the piecewise invariant in Section 3. Neither implies that every reserve claim is immediately payable in cash.

2.3 Cash, custody, and liquidity shares

Accounted cash is only one claim on the physical reserve vault. Claimable spot fees, hLP backing inventory, and protected recenter reserves also occupy custody. The program requires physical tokens to cover their sum:

$$V_i^{\mathrm{reserve}} \;\geq\; C_i + F_i^{\mathrm{custody}} + B_i^b + B_i^q + Z_i, \tag{5}$$

where B_i^b, B_i^q are backing inventory assigned to the two hLP vaults and Z_i is protected recenter funding. An unsolicited token donation does not automatically increase executable liquidity. Interest-vault accounting is separate from this reserve-vault check.

yLP is a fungible, market-wide two-sided liquidity claim. It does not give each holder an individual tick range. For total yLP supply S and a burn of s shares, the proportional reserve amounts use the stored live basis:

$$a_i = \left\lfloor \frac{sR_i}{S} \right\rfloor . \tag{6}$$

The withdrawal must still pass cash, market-health, and slippage checks. The existence of a share claim is not an unconditional redemption promise. The program checkpoints published yield before changing eligible balances; a principal withdrawal does not replace a separate harvest of that yield.

2.4 Credit Neutrality

At the principal step of a cash-backed credit draw of a, C_i decreases by a and debt increases by the corresponding indexed share amount. Ignoring rounding and simultaneous accrual, $\Delta C_i = -a$, $\Delta D_i^{\mathrm{cb}} = a$, and $\Delta R_i = 0$. This is credit neutrality: replacing inventory with a loan claim need not itself trade through the curve. Integer aggregate share conversion can produce a small reserve adjustment, and the full instruction also refreshes market state.

The cancellation can be read directly from the invariant:

$$R_i' = (C_i - a) + (D_i^{\mathrm{cb}} + a) + H_i = R_i, \qquad X_i' = R_i' - J_i = X_i. \tag{7}$$

With the other side, hLP state, and geometry fixed, the same executable point produces the same marginal price. The credit draw consumes payout capacity and changes utilization even though its principal step leaves the spot coordinate unchanged. This distinction is what allows a GAMM to lend inventory without treating a loan as a trade.

For repayment r against debt D and principal P, the principal portion is $p = \lfloor rP/D \rfloor$, with exact principal clearance on full repayment. The remainder $\iota = r - p$ is realized interest. Principal returns to reserve cash; interest follows the interest distribution path. In the idealized accounting step, $\Delta R_i = -\iota$ and $\Delta J_i = -\iota$, so X_i is unchanged. Accruing a receivable and realizing yield are different events.

For example, a quote reserve of 1,000 becomes 900 cash plus 100 debt after a principal credit draw. If the debt grows to 110, the stored reserve becomes 1,010, while unpaid interest is 10 and executable principal remains 1,000. Full repayment restores 1,000 cash and realizes 10 interest for distribution. The cash composition changes at each step; unpaid interest never becomes extra spot principal merely because time passes.

2.5 Why several reserve coordinates are necessary

Equation (1) records ownership claims, not a guarantee that all claims can leave custody at once. A public loan can remain outstanding while a trader obtains a quote, but the trader's

output still needs a cash source. Similarly, an hLP live-reserve entry has an owner and a funding obligation, yet does not create an additional token in custody. A safety check on the ledger and a safety check on the transfer therefore answer different questions.

Equation (3) removes a further ambiguity. Unpaid interest belongs in the debt balance and influences risk, rates, and share accounting. Allowing it to create executable spot principal would quote a receivable as if it were newly supplied inventory. The program explicitly subtracts public and isolated unpaid interest before constructing its executable reserve state. The hLP transition separately accounts for hLP funding interest.

The total yLP supply includes the balances owned by both aggregate hLP vaults. During an integrated spot trade, the ordinary holders' share count remains fixed while the vault-owned balances may be reconstructed. It is therefore useful to distinguish S, total supply used in proportional ownership, from S_o, the ordinary share supply held outside those vaults. Treating them as interchangeable would obscure who receives a fee or absorbs a change in vault equity.

3 Spot Execution and Yield

Trading occurs on an ordinary yLP tranche (U, V) obtained after removing hLP ownership from (X_b, X_q). Let $z = \sqrt{p}$ denote the curve's marginal quote-per-base price in normalized units. The continuous reserve path is a full-range tail of liquidity $L_0 > 0$ plus up to two nested ranges. For range j with liquidity L_j and square-root boundaries $a_j < b_j$, define $\bar{z}_j = \mathrm{clip}(z, a_j, b_j)$. Then

$$U(z) = \frac{L_0}{z} + \sum_j L_j \left(\frac{1}{\bar{z}_j} - \frac{1}{b_j} \right), \tag{8}$$

$$V(z) = L_0 z + \sum_j L_j (\bar{z}_j - a_j). \tag{9}$$

The inner range is the core; the wider range supplies the shoulder. In the core both are active. In either shoulder only the wider range is active. Outside both ranges, the full-range tail remains. The normal nested path has five regions separated by four boundaries.

Within one region, active liquidity L_A is constant. A base input ΔU moves the square-root price down according to

$$z' = \left(\frac{1}{z} + \frac{\Delta U}{L_A} \right)^{-1}, \qquad \Delta V = L_A (z - z'). \tag{10}$$

If an input reaches a boundary, the quote continues in the adjacent region. Quote input is symmetric: $z' = z + \Delta V / L_A$ and base output is $L_A(1/z - 1/z')$. This gives a bounded segment walk rather than an iterative invariant root solve. Geometry preparation still uses bounded integer arithmetic; the statement concerns the execution path across prepared segments.

The three shape controls are peak amplification, core half-width, and fade width. With a shoulder enabled, concentrated liquidity is divided between the inner and outer ranges, with integer rounding favoring the core. At amplification one and zero widths the curve is exactly

constant product. In that special case $p = V/U$ and output is $\Delta U\, V/(U + \Delta U)$. In a concentrated region p is the marginal price obtained from geometry, generally different from the inventory ratio V/U.

3.1 The curve invariant

Inside a fixed region r, equation (9) can be written as $U = L_A/z + \alpha_r$ and $V = L_A z + \beta_r$. The constants α_r, β_r collect finite-range offsets and inventory from inactive ranges. Hence the local exchange invariant is

$$\boxed{(U - \alpha_r)(V - \beta_r) = L_A^2, \qquad -\frac{dV}{dU} = z^2 = p.} \tag{11}$$

At a boundary the reserves and marginal price remain continuous, while the active set, offsets, and L_A change. In the pure CPMM case both offsets vanish and $UV = L_0^2$. This preserves the original invariant-based explanation of exchange while allowing concentration. It does not impose a constant raw product $R_b R_q$ across loans, interest accrual, fees, liquidity changes, and recentering.

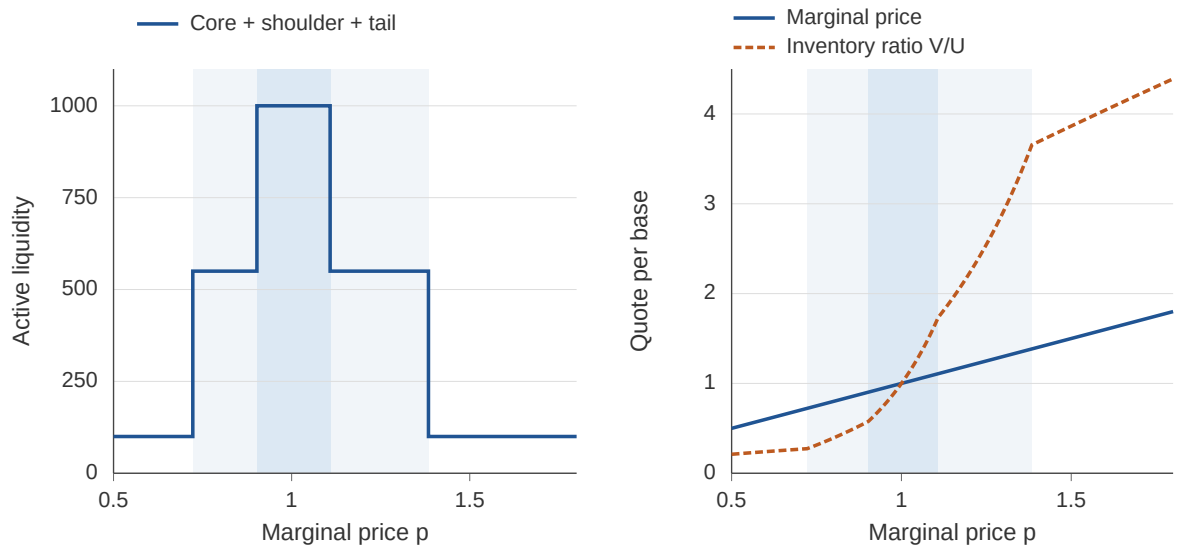


Figure 1: An illustrative fixed nested geometry. Marginal price and inventory ratio remain continuous across the boundaries, while active liquidity changes by region. The dark core and light shoulder describe the same two ranges in both panels. The example is specified in the companion data.

3.2 Center movement and fees

The concentration center is persistent state. An eligible operation can commit a previously deferred controller target before preparing its quote. The new trade observation may schedule

a later target; it does not first move the center using its own endpoint and then reprice itself against that center. Thresholds, an adjustment step, an interval, and economic admission checks constrain center movement.

The spot fee combines a base charge, an outward-divergence surcharge, and a volatility surcharge. The active path evaluates divergence over a gross trade path and limits the charge with component and aggregate budgets. It does not run the historical implicit fee solver. Volatility pricing uses decayed prior state, and a successful trade updates the signal used by later operations.

Fees can be collected on input, on base only, or on quote only. If the fee asset is input, the budget is measured on actual reserve credit after the input token transfer fee. If the fee asset is output, it is measured on gross curve output. In either case total Dusk spot fees are bounded by half of that fee basis, rounded down. Token-program transfer fees are additional; the bound is not a universal limit on the user's total loss from fees and price impact.

3.3 Claimable yield and compounding

A configured portion of eligible LP base fees and distributed dynamic surcharges can compound into principal. The remaining claimable spot yield is held outside executable reserves and distributed through growth indexes. Realized credit interest follows its own distribution path. Thus the earlier description of all yield as universally non-compounding no longer applies.

For an eligible balance s, an index increment ΔG_i produces a yield increment of the form $s\Delta G_i$, with fixed-point scaling and fractional remainder handling in the implementation. Eligibility must be measured at the correct transition boundary: changing hLP ownership before allocating a fee could assign the trade's yield to a different set of shares.

Retained dynamic surcharge has another purpose. It enters Z_i in equation (5) and remains unavailable for spot trades or harvest until an admitted recenter deploys it. It is not interchangeable with claimable yield, base-fee compounding, or unpaid interest. The controller's ability to move liquidity is therefore constrained by a specific funding source.

3.4 Launch controls and recipient authority

The fee profile includes an optional launch schedule. Its time decay can be linear or use a sixteen-step exponential schedule; an optional price-progress schedule also has a bounded duration. A directional size premium prices an individual qualifying buy. It is stateless across transactions, so splitting an order across transactions can reduce the premium. While the directional limiter is active, the program rejects multiple same-market top-level actions that execute spot trades in one transaction. These rules do not imply a universal per-wallet trading quota or resistance to all forms of order splitting.

Yield ownership, destination, and execution authority are distinct. The yield-account owner can set the recipient and an optional harvest authority independently. The owner, recipient, or configured harvest authority may invoke harvest, but payout goes to the recipient's canonical

associated token account for the mint and token program. The additional authority authorizes execution of harvest; it does not give the signer arbitrary payout routing. An owner that is a program-derived address can act through its program's signed invocation.

Compounding changes a reserve claim, whereas harvest settles a published liability. The integrated fee transition allocates value to the pre-trade eligible yLP ownership, converts each hLP's allocation into its target equity, and reconstructs the matching funding position. This is why simply adding the entire fee to one raw reserve after a trade is not a complete description of the current fee path.

4 Depth Without Spot Motion

Spot is a marginal exchange rate; depth determines how quickly that rate changes as an order consumes inventory. Two curves can quote the same infinitesimal price and deliver different outputs for a finite trade. The original radial-scaling argument captures that distinction and extends to the nested curve.

4.1 Scaling ordinary liquidity

Hold the concentration center, range boundaries, and relative liquidity weights fixed. Scale every layer, including the tail, by $\lambda > 0$. Equation (9) is linear in those liquidities, so at the same square-root price z,

$$\widetilde{L}_j = \lambda L_j, \qquad \widetilde{U}(z) = \lambda U(z), \qquad \widetilde{V}(z) = \lambda V(z), \qquad \widetilde{p} = p. \tag{12}$$

Writing $\lambda = 1 + \eta$ recovers the earlier condition $\Delta U/U = \Delta V/V = \eta$. The price conclusion follows from scaling the same geometry. For a concentrated curve it cannot be justified merely by identifying V/U with spot.

For an input a that stays within one region, scaling L_A gives

$$\widetilde{z}' = \left(\frac{1}{z} + \frac{a}{\lambda L_A}\right)^{-1}, \qquad b_\lambda = \lambda L_A(z - \widetilde{z}'). \tag{13}$$

More ordinary liquidity means a smaller price movement for the same input. Across boundaries, the required input to reach each boundary scales by the same factor. Equivalently, the full fee-free output function obeys $b_\lambda(a) = \lambda b_1(a/\lambda)$ for a fixed curve shape and starting price.

The CPMM limit makes this visible without a segment walk:

$$b_\lambda(a) = \frac{a\lambda V}{\lambda U + a}, \qquad 1 - \frac{b_\lambda(a)/a}{V/U} = \frac{a}{\lambda U + a}. \tag{14}$$

The right-hand side is the average execution-price shortfall from the initial marginal price, before fees. For $U = V = 1{,}000$ and a fixed 50-base input:

Scale λ	Ordinary reserves	Quote output	Shortfall (bps)
1	(1,000, 1,000)	47.61905	476.19
2	(2,000, 2,000)	48.78049	243.90
4	(4,000, 4,000)	49.38272	123.46

All three rows start at one quote per base. Their outputs differ because scaling the ordinary tranche changes the exchange path. These calculations assume sufficient output cash; equation (1) permits a reserve to contain debt, so depth and immediately payable cash still need separate checks.

4.2 Scaling hLP ownership claims

The hLP endpoint has a related radial identity with a different role. For target equities E_b, E_q, define

$$\omega = 1 + \frac{E_b}{U} + \frac{E_q}{V}. \tag{15}$$

The canonical reconstruction in Section 5 then gives

$$(X_b, X_q) = \omega(U, V), \qquad \frac{X_q}{X_b} = \frac{V}{U}. \tag{16}$$

This explains how a larger gross reserve coordinate can coexist with the same spot: at fixed (U, V), an ownership reconstruction changes the scale of the total claims without moving the ordinary curve point. In a concentrated market the common inventory ratio still differs in general from marginal price.

The quote is evaluated on (U, V) before reconstructing hLP claims. Holding that ordinary tranche fixed while increasing ω therefore does not grant a trader the larger output in equation (14). For example, $E_b = 100$ and $E_q = 50$ around $(1{,}000, 1{,}000)$ give $(X_b, X_q) = (1{,}150, 1{,}150)$. A 50-base CPMM input still receives 47.61905 quote before fees, because the ordinary tranche remains $(1{,}000, 1{,}000)$. Treating the gross claims as the spot reserves would instead quote 47.91667.

Credit neutrality, ordinary depth scaling, and hLP reconstruction thus describe three distinct transitions. A principal credit draw exchanges cash for debt; proportional ordinary liquidity changes the scale of executable depth; hLP reconstruction changes ownership and funding around the executable point. The GAMM invariant accounts for each, while the quote path determines which transition actually changes trade impact.

5 Funded Liquidity (hLP)

5.1 From Yield Basis's square-root intuition to Dusk's hedge

The constant-product starting point is the leveraged-liquidity argument discussed by Egorov and Yield Basis [5, 7]. The following steps derive the ideal benchmark; Dusk's finite transaction accounting is described afterward.

Step 1: solve the ordinary LP inventory. Let x be META, y be USDC, and p the USDC price of META. For a fixed, fee-free constant-product LP slice,

$$xy = k, \qquad p = \frac{y}{x} \quad \Longrightarrow \quad x = \sqrt{\frac{k}{p}}, \qquad y = \sqrt{kp}.$$

Its total USDC value is therefore

$$W(p) = px + y = 2\sqrt{kp}, \qquad \frac{W(p)}{W(p_0)} = \sqrt{\frac{p}{p_0}}, \qquad \frac{dW}{W} = \frac{1}{2}\frac{dp}{p}. \tag{17}$$

For a small one-percent price move, this LP's value moves approximately half a percent before fees. The factor one half comes from its inventory, rather than from a selected risk parameter.

Step 2: distinguish static borrowing from maintained leverage. At META price $p_0 = 1$ USDC, deposit equity worth 100 USDC, borrow 100 USDC, and hold 200 USDC of LP assets. Gross leverage is $200/100 = 2$. For unchanged debt and price ratio $r = p/p_0$, equity becomes

$$E_{\text{static}}(r) = E_0(2\sqrt{r} - 1), \qquad E_{\text{hold}}(r) - E_{\text{static}}(r) = E_0(\sqrt{r} - 1)^2. \tag{18}$$

At $r = 0.81$, the LP assets are worth 180, debt is still 100, and equity is 80. Holding the original 100 META is worth 81. Borrowing once therefore leaves a one-unit tracking shortfall in this CPMM example.

Step 3: derive the ideal maintained-2× result. Let n be the number of LP units, W the value per unit, and $E = nW - D$ equity after quote debt D. A costless self-financing rebalance changes debt and LP units together: $dD = W\,dn$. Consequently $dE = n\,dW$, giving

$$\frac{dE}{E} = \underbrace{\frac{nW}{E}}_{\mathcal{L}} \frac{dW}{W}.$$

If $\mathcal{L} = 2$ is continuously maintained along a smooth price path,

$$\frac{dE}{E} = 2\left(\frac{1}{2}\frac{dp}{p}\right) = \frac{dp}{p} \quad \Longrightarrow \quad E(p) = E_0\frac{p}{p_0}.$$

This benchmark assumes no funding cost, fees, execution loss, or discrete jumps. Yield Basis uses Curve collateral and an arbitrage-driven maintenance mechanism; its rebalance economics must be accounted for separately. The derivation does not prove costless finite execution for Dusk or for every concentrated pool.

Step 4: replace the one-half weight under concentration. For a fixed, fee-free differentiable curve, the marginal price satisfies $p\,dx + dy = 0$. Differentiating $W = px + y$ gives

$$dW = x\,dp + p\,dx + dy = x\,dp, \qquad \frac{dW}{W} = \underbrace{\frac{px}{px + y}}_{w_b}\frac{dp}{p}.$$

To obtain unit local META sensitivity with quote funding, the required gross leverage is

$$\boxed{\mathcal{L}_b = \frac{1}{w_b} = 1 + \frac{y}{px}.} \tag{19}$$

With base funding and quote-target equity, the symmetric expression is $\mathcal{L}_q = 1 + px/y$. This local statement holds inside a fixed region; changes of geometry, cash flows, and finite rebalancing require their own accounting.

META share of gross LP value	Gross META hedge leverage
50%	2×
62.5%	1.6×
40%	2.5×

In plain language, borrow against the opposite-token claim until it is canceled, leaving the target-token claim. A price move changes inventory weights under concentration even if the configured amplification stays fixed. At an arbitrage-aligned CPMM point $px = y$, the weight stays one half and the ratio stays two. Dusk's target is the matching opposite claim and debt; the resulting gross leverage follows the inventory weights.

5.2 A funded position inside the GAMM

Yield Basis develops leveraged LP exposure around Curve Cryptoswap collateral and crvUSD debt, with a LEVAMM and VirtualPool arbitrage route maintaining the position [6]. Dusk puts the liquidity shares, funding debt, and exchange path in the same pair-local accounting system. It can therefore reconstruct the funded ownership at the endpoint already produced by its own quote.

The base hLP vault holds a funded yLP position whose net exposure targets base; quote hLP is symmetric. Let E_b, E_q be their target-asset equities after accounting for the funding and yield effects included in the transition. With no unpaid public interest or book offsets, the continuous canonical endpoint is

$$X_b = U + E_b + E_q\frac{U}{V}, \qquad D_b^{h,*} = E_q\frac{U}{V}, \tag{20}$$

$$X_q = V + E_q + E_b\frac{V}{U}, \qquad D_q^{h,*} = E_b\frac{V}{U}. \tag{21}$$

The opposite-asset yLP claim and the target funding obligation match. For ordinary share supply S_o, vault-owned shares are reconstructed as

$$s_b = \left\lfloor \frac{S_o E_b}{U} \right\rfloor, \qquad s_q = \left\lfloor \frac{S_o E_q}{V} \right\rfloor, \qquad S = S_o + s_b + s_q. \tag{22}$$

These formulas describe aggregate vault state and require no scan over hLP holders.

For base hLP, the fraction E_b/U of the ordinary portfolio supplies base claim E_b and quote claim $E_b V/U$. Matching the latter with quote debt leaves the target base claim. Quote hLP does the converse. At a canonical settled endpoint, the ledger's funded entries are $H_b = D_b^{h,*}$ and $H_q = D_q^{h,*}$; the complete hLP-owned inventory additionally includes the target equities. Funding interest can make actual indexed debt differ from these canonical targets before settlement or recovery.

The trader first receives a quote on (U, V). The program then reconstructs the two vaults at the quoted endpoint and verifies the associated debt, cash, interest, and reserve changes. Synthetic hLP inventory is not an extra free tranche of trader depth. The earlier fixed-point pre-rebalance, bisection, and tracking-loss-threshold description is not the active integrated mechanism.

For base hLP, its gross LP value at marginal price p is $E_b p + E_b V/U$, and debt value is $E_b V/U$. Relative to target equity value $E_b p$, gross leverage is

$$\mathcal{L}_b = 1 + \frac{V}{Up}, \qquad \mathcal{L}_q = 1 + \frac{Up}{V}. \tag{23}$$

Both equal two at any arbitrage-aligned constant-product point, where $V/U = p$. With concentrated liquidity away from the center, inventory weights differ and a universal fixed-2× claim would be incorrect. The target is one-sided net inventory, and the leverage needed to represent it follows the curve's actual inventory weights.

5.3 Equity, NAV, and the meaning of the hedge

For base hLP at a canonical endpoint, quote-denominated NAV is

$$\mathrm{NAV}_b = pE_b + E_b\frac{V}{U} - D_q^{h,*} = pE_b. \tag{24}$$

This is an explicit asset-minus-debt identity: the opposite claim cancels the opposite debt. It is stronger than saying that gross inventory was multiplied by two, because it specifies what is owned, what is owed, and which asset remains after netting. With $U = V = 1{,}000$, $p = 1$, and $E_b = 100$, the vault owns 100 base plus 100 quote and owes 100 quote; its NAV is 100 quote, representing 100 base.

Equity is determined from the current reserve claims and indexed debt when the transition is prepared, rather than assumed from a stale NAV checkpoint. The canonical formula then describes the reconstructed state. Between endpoints, funding interest, fees, recovery payments, losses, and rounding can change the equity available to reconstruct. At fixed target equity the neutral endpoint follows the target asset; the protocol does not promise to hold that equity fixed through every operation.

The hedge therefore has a precise scope: cancel the funded opposite-asset claim at an admitted endpoint while retaining target-asset and LP-fee exposure. It does not hedge the target asset's price, remove funding cost, or guarantee a cash exit. Positive marked NAV is necessary for ordinary solvent operation, but the settlement conditions below also determine whether that claim can be paid.

5.4 Funding and settlement constraints

An hLP deposit is admitted only if projected indexed funding debt fits the current borrowed-asset cash headroom. This is a deposit check, not a global theorem that funding debt remains below cash after every swap or after passive interest accrual. A spot swap first quotes full

accrued hLP funding-interest settlement. If that payment would consume cash needed for trader output or the final reserve atom, it requotes without a funding-recovery bonus and carries both active vaults' unpaid funding interest on indexed debt. The hLP debt principal and synthetic borrowed reserve remain separate, so carrying interest cannot erase the liability or create spendable AMM principal. A later swap can pay the carried interest when cash is sufficient. Trader output must still be covered by actual reserve cash; other transition types retain their own cash requirements.

The implementation binds a prepared hLP transition to its expected pre-state. After applying the cash and funding effects, it checks the reconstructed executable endpoint within a fixed tolerance of three raw token atoms per side and asserts the stored reserve identity. Token normalization, indexed debt conversion, and share rounding remain part of that check. The zero residual-exposure field is an accounting result of reconstruction, not a guarantee against economic loss.

5.5 Funding recovery and terminal closeout

Let D be actual indexed funding debt, A the canonical opposite claim, and Y revenue already available for netting. The recovery gap is

$$g = [D - Y - A]_+ . \tag{25}$$

For positive A, recovery ignores gaps below 25 basis points of A. Above that threshold the discount is

$$d = \min\left(0.05, \frac{0.8g}{A}\right). \tag{26}$$

It reaches its maximum at $g = A/16$. A spot trade supplying the borrowed asset can match $m = \min(g, \text{input})$. If that matched amount converts to a fair target amount v, the extra target output is

$$\beta = \min\left(E_t, \left\lfloor \frac{vd}{1-d} \right\rfloor\right). \tag{27}$$

The full trade still pays ordinary fees and follows the ordinary curve. The bonus consumes the selected hLP's equity. Critical stress is reported on actual debt before revenue netting, at $D/A \geq 9/8$; it does not gate ordinary swap recovery. Swaps remain available in reduce-only mode and require the caller's input. The atomic flash recovery accepts a positive funding gap, or a frozen vault, before that boundary. It fixes a net opposite-token payment, transfers target collateral at an affordable discount, and requires same-transaction settlement. A full flash close remains subject to the terminal insolvency checks.

Terminal hLP closeout is narrower than arbitrary debt forgiveness. It requires insolvency at both the executable curve mark and symmetric EMA, retires the vault's yLP exposure, and allocates recovery to principal first. Bounded insurance covers eligible principal shortfall; remaining principal loss is written off and unpaid interest is canceled. Previously funded holder yield claims remain recorded. This is separate from ordinary borrower liquidation and from a routine hLP withdrawal.

5.6 Leveraged liquidity and Yield Basis

The constant-product LP convexity argument explains why two-times leverage is a useful reference point [5]. It does not prove that every concentrated LP portfolio has that leverage, or that funding and trading losses cancel. Yield Basis maintains leverage around externally held Curve LP collateral and crvUSD debt [6]. Dusk instead matches opposite-asset claims and debt inside a pair-local endpoint reconstruction. Neither architectural description establishes a ranking of returns, costs, or resilience.

Dusk's target-asset equity can change through funding interest, fee allocation, recovery incentives, settlement, and market losses. A neutral opposite-asset claim at an admitted endpoint removes one accounting mismatch. It does not make target equity constant across every operation. In particular, equation (27) makes the cost of attracting recovery flow explicit: the trader's additional output is a reduction in hLP equity.

6 Credit, Rates, and Liquidation

6.1 Credit admission

Public credit uses a shadow constant-product risk curve derived from the full-range tail, conservative depth, and internal price references. In a collateral-to-debt orientation, write

$$\widehat{p} = \min(p_{\text{symEMA}}, p_{\text{dirEMA}}), \qquad L_{\text{hist}} = \min(L_{\text{observed}}, L_{\text{EMA}}). \tag{28}$$

When one historical depth observation is zero, the nonzero observation is used. Write $L_{\text{total}} = L_0 + L_{\text{concentrated}}$ for current prepared center depth. The shadow liquidity scales only the tail:

$$\widehat{L}_0 = L_0 \frac{\min(L_{\text{hist}}, L_{\text{total}})}{L_{\text{total}}}, \tag{29}$$

with $\widehat{L}_0 = L_0$ when historical depth is unavailable. The conceptual risk reserves are $\widehat{x} = \widehat{L}_0/\sqrt{\widehat{p}}$ and $\widehat{y} = \widehat{L}_0\sqrt{\widehat{p}}$. Impact-aware value is

$$W(c) = \frac{c\widehat{y}}{\widehat{x} + c}. \tag{30}$$

The full concentrated execution curve is not simply substituted into credit admission. The public credit book also maintains debt-capped aggregate underwriting contributions; those aggregate contributions do not lock other accounts' idle collateral.

A successful public credit draw recomputes and stores the position's liquidation collateral factor, subject to the borrower's minimum acceptable factor. Credit capacity uses a five-percent buffer below that factor. On collateral withdrawal the check uses the greater of the stored and newly computed factor, then tests the post-withdrawal buffered capacity. Full repayment clears the factor. Each debt side is secured by that position's opposite-asset collateral.

Public credit draws additionally consume a leaky 24-hour budget derived from conservative depth. The default limit is 20 percent and the configurable ceiling is 30 percent of the relevant

depth measure. Only new public credit principal consumes this budget; repayment does not refill it immediately. hLP funding and isolated margin do not consume this particular public credit bucket, although both face their own checks and enter utilization.

6.2 Adaptive interest

For each asset, utilization includes all indexed debt without double counting:

$$u_i = \frac{D_i^f + D_i^\ell + D_i^h}{D_i^f + D_i^\ell + D_i^h + C_i}. \tag{31}$$

The default target is $u^* = 0.70$, with normalized error

$$e(u) = \begin{cases} (u - u^*)/u^*, & u \le u^*, \\ (u - u^*)/(1 - u^*), & u > u^*. \end{cases} \tag{32}$$

For curve steepness κ, the instantaneous APR is $r = r_T c(e)$, where

$$c(e) = \begin{cases} 1 + (\kappa - 1)e, & e \ge 0, \\ 1 + (1 - 1/\kappa)e, & e < 0. \end{cases} \tag{33}$$

The default $\kappa = 4$ gives multipliers from 1/4 to 4. Dusk uses the adaptive-anchor idea associated with Morpho's AdaptiveCurveIRM [4], with a bounded linear update

$$r_T' = \mathrm{clip}\left(r_T\left[1 + \mathrm{clip}(se\tau, -0.5, 0.5)\right], 0.001, 2\right), \tag{34}$$

where τ is elapsed time in years, capped at one year per update, and default speed is $s = 20$. The anchor starts at four percent APR. Its 0.1–200 percent bounds apply to the anchor, not the final debt APR after multiplication.

The program accrues the debt index at the rate computed from the pre-update state, then updates the anchor. Elapsed time comes from the difference between stored and current `Clock.unix_timestamp`, converted from seconds to milliseconds. Repeated timestamps accrue no time; slots retain observation ordering and atomic-session checks. Debt and index arithmetic round in integer fixed point, so continuous-time formulas or a particular update cadence must not be presented as bit-for-bit execution under arbitrary activity.

6.3 Immediate flash liquidation

Public credit eligibility compares indexed debt with a linear collateral mark at the symmetric EMA and the position's stored liquidation factor. Margin positions compare symmetric-EMA equity with their stored maintenance requirement. Current AMM execution alone cannot liquidate an EMA-healthy position. Token-fee notice rules can impose additional conservative valuation before an announced fee activates.

The public auction instructions have been replaced by immediate flash liquidation and emergency settlement. A keeper may pay from its own funds or atomically receive a permitted

collateral slice, exchange it through a route of its choice, and settle the bound debt-token obligation. Dusk does not require a particular Jupiter route. Ordinary Dusk swaps can participate subject to real output-cash constraints. A swap followed by repayment can require more intermediate cash than an integrated debt offset.

The begin instruction fixes the collateral debit, actual net debt repayment, charge allocation, and session identity; settlement must fulfill those obligations in the same transaction. Position locks prevent conflicting changes to that position. Unrelated market actions remain possible subject to reservations and current-state accounting. Settlement cannot restore a stale market snapshot or count a fee payment twice as repayment. A missing or underfunded settlement causes atomic rollback. SDK builders and previews bind the complete transaction and measured net token credits.

Let $e = (V - D)/V$ be EMA equity fraction and m the maintenance fraction. Define distress progress $h = \mathrm{clip}((m-e)/m, 0, 1)$ and $t = \min(a/120, 1)$ for the age a in seconds of a committed distress episode. The ordinary buyer discount ceiling is

$$\delta_{\max} = 0.005 + 0.045 \max(h, t). \tag{35}$$

Fills are available immediately; 120 seconds is an incentive ramp, not a waiting period or expiry. A smaller partial-fill discount, down to 0.5%, is chosen when required to improve surviving health after all costs. Useful partial fills do not reset continuing distress. Verified recovery clears the episode. A failed transaction cannot establish a persistent timer.

For net debt repayment R, the solvent liquidation charge is $F = \lfloor R/100 \rfloor$, subject to the full-settlement surplus cap. The buyer funds $R + F$ after transfer fees; F is not deducted from the advertised R. Before token-fee adjustments, a collateral amount c at reference price p satisfies $cp(1 - \delta) = R + F$. The buyer keeps execution upside. For $R = 1{,}000$, payment is 1,010, of which 1,000 repays debt, 2 goes to protocol and 8 to the insurance/LP allocation. The protocol cannot create a shortfall merely to collect the charge.

Partial fills must improve the surviving position after rounding, transfer costs, incentives and charges. The recovery target is surviving maintenance plus two percentage points; useful smaller improvements remain possible. There is no creator-configured minimum remaining debt and no fixed insurance-funded cleanup bonus. A position can remain unfilled if every available route is uneconomic.

6.4 EMA-priced emergency settlement and losses

Emergency access requires EMA equity at or below 70% of maintenance, including nonpositive equity. Time alone never opens this permission. At 7% maintenance, the threshold is 4.9% equity. The caller reward is a fraction of proceeds,

$$b = \max(0.01h,\ 0.003 + 0.007t) \le 0.01,$$

including insolvent execution. The ordinary buyer discount and this caller reward are different incentives.

Internal execution reconstructs a reference curve from the collateral-side symmetric EMA, smoothed curve depth and configured concentration. The reference prices the collateral-for-debt offset; reference reserves are never installed as live inventory. Seized collateral enters real pool reserves, the corresponding debt is resolved, and real curve/hLP accounting is rebuilt around the resulting inventory. Normal configured swap fees remain applicable. Reference-curve size impact still exists: this is not a flat EMA multiplication or a zero-slippage promise. Physical rewards, fees, funded interest and owner payouts still require actual cash.

Emergency partials are preferred when useful. Inside the emergency gate, full internal closure is authorized for nonpositive EMA equity or when the program conservatively proves that no smaller permitted internal settlement can restore maintenance plus two points after costs. A below-target improving partial or a better external route does not veto that particular full-internal fallback. Ordinary flash full closure uses its stricter no-useful-partial certificate. A keeper-selected failing size or an exhausted proof budget is not proof of infeasibility.

Recovery protects principal first. Net proceeds repay principal, then any payable interest; insurance covers only the eligible remaining borrower/margin principal shortfall, and unpaid interest is canceled. Residual principal is written off to the market. No protocol/referral yield is manufactured from canceled interest, and no additional treasury liquidation charge is funded by insurance or writeoff. For principal 900, interest 50 and net recovery 850, the principal gap is 50. With sufficient budget, 75% coverage supplies 37.5, principal writeoff is 12.5 and the unpaid 50 interest is canceled.

Insurance targets are per debt token:

$$T = \lceil P/20 \rceil + \lceil H/15 \rceil ,$$

where P is covered public/margin principal and H is indexed hLP funding debt. Of the collected solvent charge, 20% is protocol revenue and 80% is designated for insurance/LPs. This designated portion funds insurance fully through 75% of target, tapers linearly to zero at target, and sends the remainder to existing LP yield ownership, including hLP through yLP. Existing insurance is not paid out merely because its target falls.

Borrower/margin coverage is at most 75% of eligible principal loss, further bounded by available funds and configured limits. Eligible hLP terminal shortfalls retain their separate coverage rule. All claimants share the fund's at-most-50% spending budget, which refreshes after 24 hours of elapsed on-chain timestamp time using the remaining balance. Changing claimant type does not reset that budget.

The original live-price/depth dependence in finding #287652 is remediated. The owner accepted the remaining inventory/arbitrage risk: after the reference exchange, LP inventory can lose value through subsequent trading. Such inventory loss is not a second outstanding borrower principal claim that automatically draws insurance. This acceptance does not cover a new accounting defect or establish best execution, absence of manipulation profit, or universal liquidation liveness.

6.5 Stored margin terms and position actions

Each margin position holds one asset and owes the opposite asset. Entry accepts the debt token only: a META long adds USDC, borrows USDC and buys META; the reverse adds META, borrows META and sells it for USDC. Either token may be volatile. A UI zap can convert another wallet asset before entry. Choosing less leverage changes funding amounts, not the protocol's maintenance formula.

Maintenance applies progressive marginal rates of 7/12/17% at boundaries of 5/15% of a stored collateral-side depth reference L. Writing N in the same units,

$$M(N, L) = .07 \min(N, .05L) + .12 \min((N - .05L)_+, .10L) + .17(N - .15L)_+.$$

Initial equity must cover effective maintenance plus three percentage points and the retained admission obligation. At $N = .30L$, maintenance is about 13.67% and the size-only initial rate is about 16.67%, approximately a six-times ceiling before crowding and execution costs. The old independent two-percent unwind-impact rejection is removed; the requested twenty-times multiplier remains an outer bound.

Aggregate directional exposure affects new admission through a cumulative equity potential. Let $S(A, L)$ apply marginal 10/15/20% initial rates at the same boundaries to aggregate exposure A; set

$$C(A, L) = A\big(.10 + \max(0, A/L - .20)/10\big), \qquad F(A, L) = \max(S, C).$$

A risk increase retains the positive change in F across its pre/post states, in addition to its own size requirement. At unchanged depth, sequential slices cannot avoid that aggregate charge: at $L = 50{,}000$, one 45,000 exposure and three successive 15,000 exposures both retain 8,500. Distinct positions can still have different maintenance behavior. Depth uses conservative cash and smoothed observations; synthetic inventory and unpaid receivables do not create spendable capacity.

Existing terms do not change just because another trader enters or LP liquidity changes. Price and interest still change actual health. A risk increase rechecks the whole position and may tighten stored depth; repayment does not reprice terms. Proportional closure releases the corresponding retained obligation. Debt-free collateral is removed from crowded exposure; borrowing again re-admits the entire resulting position.

Increase-size borrows and buys more held tokens. Decrease-size sells held tokens to repay debt. Increase-margin repays debt while retaining held-token quantity. Add-to-position combines wallet funding and optional borrowing, targeting whole-position leverage. Remove-from-position proportionally sells collateral and repays debt, returning net residual to the owner. Standalone debt extraction/decrease-margin is unsupported. Repayment or reduction can reach a debt-free 1× position; initial margin entry still requires borrowing. SDK sizing helpers require transaction simulation for final admission and net wallet flows.

6.6 Governance, delegated orders, and token admission

Parameter governance exposes seven families: fees, concentration, interest, EMA half-lives, public daily credit limits, the center controller, and insurance. A reusable owner/market gov-

ernance balance holds virtual yLP and preserves its yield. One revocable delegate may create and support proposals with its available balance; the owner retains control of withdrawals and delegate assignment. Allocating support reserves balance rather than spending it twice. Both direct and reusable support permit partial withdrawal/reassignment while collecting; queued commitments remain locked until the applicable terminal condition. Revoking delegation does not erase an existing queued commitment. Released reusable support returns to the balance, while yield stays with its owner. Proposal digests, permanent identity tombstones and original rent-refund destinations prevent reused-identity and delegation-redirection errors.

The separate order-delegate program authorizes conditional margin entries/exits and hLP stops. Dusk performs delegated close fee splitting and sends the owner's net residual to an owner-owned token account. Protocol/keeper charges and Token-2022 fees are included in net-payout checks. Conditional authority does not reserve cash or ensure timely execution. hLP order escrow preserves the owner's yield entitlement, with cleanup of funded claims after execution/cancellation where applicable.

Market admission rejects freeze-authority assets and unsupported transfer-hook behavior. Mutable transfer-fee assets remain supported: admission conservatively evaluates current and scheduled fees per amount, health anticipates an increase in its final notice epoch, and actual settlement uses the effective schedule and measured spendable credit. Fee authority is not required to be revoked. Timely keeper rescan at epoch boundaries is necessary; a late or uneconomic fill can still leave losses. LP metadata is initialized atomically with market creation using canonical formatting and caller-supplied asset labels; labels are not authenticated mint symbols. Versioned transactions and an address lookup table support the three metadata records. Program, SDK, keeper, indexer and UI versions must agree on the resulting accounts, events and net-amount conventions.

7 Accrued LP Interest and hLP Recovery

The integrated candidate prices ordinary yLP entry and burn using full accrued public-interest value. Both hLP vaults use the same full-value basis for entry, NAV and hedge reconstruction. An entrant buys its share of the existing receivable; it does not receive old interest for free. Unpaid public interest remains outside executable swap reserves, and exits pay only available cash. A holder who exits does not retain a detached claim on future repayment. Repayment funds the receivable once; writeoff cancels unpaid value. Funded holder claims remain distinct from new yield.

For a quoted trade, ordinary trading coordinates (U, V) differ from ordinary book claims $(A, B) = (U + o_b, V + o_q)$. The book offsets o_b, o_q are fixed during that quote. Opposite debt is reconstructed from $E_b B/A$ and $E_q A/B$, while the executable trading coordinate excludes unpaid public interest. At zero offsets these expressions reduce to the earlier endpoint equations. The implementation certifies the reconstructed ownership and debt after pricing, including concentrated-curve fee allocation; this identity alone does not promise cash settlement at every state.

Cash-funded interest uses a per-asset advance ledger. The LP/protocol allocation is frozen when reserve cash funds interest, and later borrower repayment clears the advance without a second payout. A harvest transfers only available cash and does not itself move swap reserves. Newly accrued hLP-owned interest first covers that vault's funding gap; only the remainder becomes holder yield. An accompanying vault-wide hedge adjustment may change AMM reserves and price. If a spot swap lacks the cash to pay hLP funding interest while honoring trader output and the reserve floor, it requotes without the recovery bonus and carries the indexed interest liability for later payment. Real trader output still needs cash.

The recovery target is matching opposite-token backing and debt; concentration can make gross leverage differ from literal 2×. Ordinary borrowed-asset swaps can earn a bounded hLP recovery bonus whenever a qualifying funding gap exists. The separate critical ratio $D/A \geq 9/8$ is diagnostic, not an extra swap gate. Atomic external recovery instead uses `preview_hlp_flash`, `begin_hlp_flash` and `settle_hlp_flash`: target collateral is a temporary receivable until the quoted opposite-token payment arrives in the same transaction. The buyer may use inventory or an external route. Full terminal settlement requires its separate insolvency and opt-in checks, with principal-first recovery, bounded insurance, explicit write-off, canceled unpaid interest and preservation of already funded holder yield. The dedicated `rescue_hlp` instruction is removed; native recovery remains available through `swap`.

This integration candidate has native and instruction-level coverage for concentration, both hLP vaults, cash funding, repayment, writeoff, transfer fees, flash rollback and spot-cash carry. It still requires a final combined-branch audit, economic review and route-specific compute simulation. The retained numerical replay below identifies its historical source snapshot separately from these current implementation claims. Nothing in this paper authorizes deployment or describes a production migration.

8 Numerical Illustrations and Limitations

The companion generator supplies the depth comparison in Section 4 and three further illustrations of the mechanism. It produces the tables and vector figures used by the two drafts. Inputs are disclosed in the generated metadata. The curve illustration holds a chosen geometry fixed; the rate illustration updates once per day at fixed utilization; recovery examples use the integer discount and bonus rules. Checks also cover the ledger transitions, radial ownership identity, and LP tracking benchmark derived above. These mathematical illustrations are not a Solana emulator. The separate native replay in the expanded appendix retains an earlier #45 benchmark snapshot for its inventory examples; token CPIs, current hLP fee conversion and validator execution remain outside that replay.

Fixed utilization	Day-30 anchor APR	Day-30 debt APR
0%	0.7376%	0.1844%
70%	4.0000%	4.0000%
100%	19.8196%	79.2786%

The rate rows use $u^* = 70\%$, $\kappa = 4$, $s = 20$, and an initial four-percent anchor. The concentrated example uses a positive tail and equal core/shoulder layer liquidities to display the five regions. It is an illustrative geometry, not a recommended deployment configuration. Recovery figures distinguish the 25-basis-point activation threshold, maximum discount at a gap of 1/16, and the separate critical test on actual debt.

Several limitations follow from the model. Cash exhaustion can prevent an otherwise valid quote or withdrawal from settling. Internal price filters can lag or be manipulated, and a positive full-range tail does not guarantee useful size at every price. Funding growth can erode hLP equity even when opposite claims are reconstructed correctly. Liquidator participation, rescue flow, and permissionless terminal calls remain liveness assumptions. Insurance and explicit loss accounting determine how losses are allocated; they do not supply unlimited capital.

The implementation also has finite arithmetic precision, state-dependent fees, and operation ordering that a continuous derivation cannot certify. The most useful next empirical work is exact-program replay across cash stress, boundary crossings, funding recovery, and liquidation, with fee and funding costs included. Historical prototype benchmarks should not be used as measurements of this source snapshot. This revision reports no throughput, return, or comparative performance result.

Reproducibility

Both papers share the same mechanism source so that corrections to equations and parameters remain consistent. `paper/simulations/generate.py` generates the numerical data, summary table, and vector figures. `paper/SOURCES.md` maps claims to implementation paths and describes the verification scope. The expanded appendices provide derivations, worked examples, and a transition-level review map. Code remains the authority for rounding, account validation, instruction ordering, and rejection conditions.

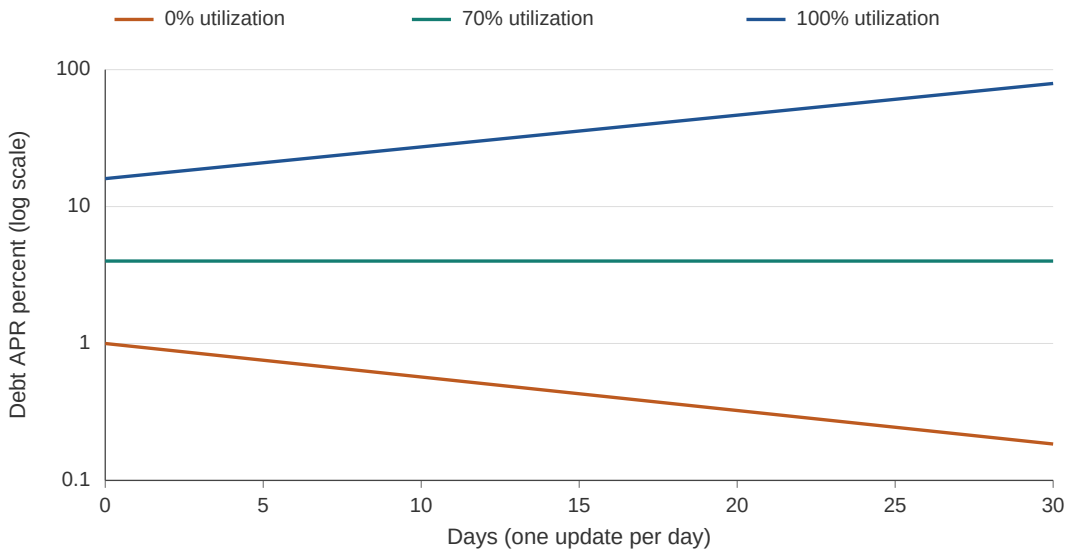


Figure 2: Fixed-utilization rate paths with one anchor update per day and current default parameters. Values show the instantaneous APR after each update. This cadence is part of the experiment, not a claim of cadence-independent onchain accrual.

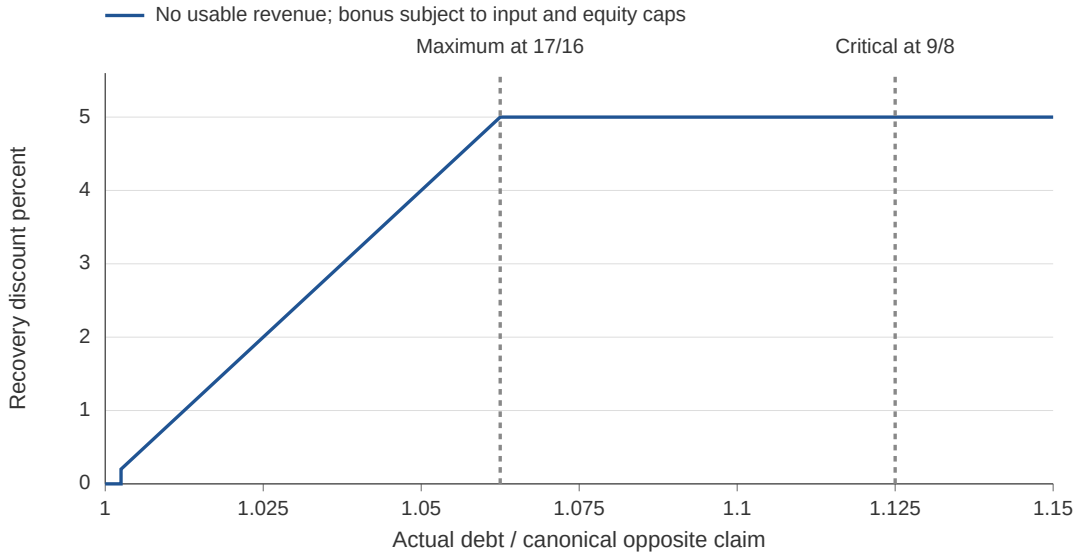


Figure 3: Recovery discount with no usable revenue. Maximum discount occurs at debt/claim 17/16; critical stress begins at 9/8. The plotted fine-scale discontinuity at 25 basis points reflects the activation guard. Bonuses also depend on matching input, conversion ratio, integer rounding, and available equity.

A Reserve Coordinates and Worked Accounting

The following notation distinguishes five objects that were collapsed in the earlier constant-product draft.

Object	Role
V_i^{reserve}	Physical token balance; custody for several separately owned buckets.
C_i	Accounted reserve cash; subject to instruction-specific availability checks.
R_i	Stored live reserve satisfying the cash/debt/hLP identity; basis of proportional yLP mint and burn amounts.
$X_i = R_i - J_i$	Executable reserve coordinate after removing unpaid public and isolated interest.
(U, V)	Ordinary yLP tranche after separating both hLP ownership claims; the trader-facing curve coordinates.

Neither the physical vault balance nor the stored live ratio is sufficient to reconstruct a concentrated spot quote. A quote also requires curve geometry, the ordinary tranche, applicable fees, cash policy, and the prepared hLP transition. This does not make equation (1) optional: every successful instruction must still account for the resulting stored reserve.

A.1 A public loan without hLP

Start with a constant-product market whose two accounted cash reserves are 1,000, with no debt, hLP, or unpaid interest. Let the quote-side borrower take 100. Ignoring share rounding, the quote-side ledger becomes

$$C_q = 900, \quad D_q^f = 100, \quad P_q^f = 100, \quad R_q = X_q = 1{,}000.$$

Base remains unchanged. The principal credit draw has changed cash composition without changing the curve point. Utilization is now $100/(900 + 100) = 10\%$ on the quote side, before any other debt or cash changes.

Suppose quote debt later accrues to 110. The ledger becomes $R_q = 1{,}010$, but $J_q = 10$ and $X_q = 1{,}000$. A quote made from raw R_q would count ten unpaid interest tokens as new curve principal. The implementation's subtraction prevents that error. Actual token cash remains 900 until another operation moves tokens.

A full repayment transfers 110 to the settlement path. The principal portion 100 replenishes C_q; the interest portion 10 follows interest distribution. Afterwards $D_q^f = P_q^f = J_q = 0$, $C_q = R_q = X_q = 1{,}000$. There is now realized yield, but it should not be added a second time to executable principal. Any later harvest settles its own published liability.

This example is intentionally independent of hLP. Once hLP is present, the same debt/principal distinction remains, while the integrated transition separately updates vault funding, ownership, and interest. Cash-backed interest and hLP funding interest must not be combined into one undifferentiated reserve increment.

A.2 A spot trade after the credit draw

Return to the state immediately after the 100 quote credit draw. A fee-free input of 100 base into the ordinary constant-product curve gives

$$\Delta q = \frac{100 \cdot 1{,}000}{1{,}000 + 100} = 90.90909\ldots.$$

The ordinary and executable reserves become $(1{,}100, 909.09091\ldots)$. Quote cash is $809.09091\ldots$, while quote debt remains 100. Thus $R_q = C_q + D_q^f$ still holds. A larger theoretical curve output would remain subject to the actual cash available for its payout.

If a fee is withheld, its destination matters. A claimable fee stays in custody outside executable reserves; a compounded LP fee enters principal through its allocation transition; retained dynamic surcharge becomes protected recenter funding. A token transfer fee can make actual reserve credit smaller than the user's gross debit. Each of these changes a different term in the accounting, so a single formula of the form "vault balance equals AMM reserve" is insufficient.

A.3 Published yield and share changes

For a growth index stored at scale 2^{64}, a user's incremental whole-token liability has the form

$$\left\lfloor \frac{s(G - G_0) + \rho}{2^{64}} \right\rfloor,$$

where ρ represents the carried fractional contribution under the relevant accounting helper. Global distribution and account-level checkpointing have distinct remainders. The point is to preserve previously earned fractions and allocate newly distributed value to the eligible supply; it is not to promise that each intermediate floor disappears.

When a holder burns principal shares, prior published yield is checkpointed before its balance changes. After terminal hLP closeout, remaining hLP tokens may represent zero principal while already published yield remains claimable. A preview or accounting dashboard should report these components separately. A single aggregate NAV number can otherwise hide a surviving liability or imply an unavailable cash payout.

B The Nested Curve as a Piecewise Analytic Path

For one bounded range $[a, b]$ in square-root-price space, its inventory is

$$x_{[a,b]}(z) = \begin{cases} L(1/a - 1/b), & z \le a, \\ L(1/z - 1/b), & a < z < b, \\ 0, & z \ge b, \end{cases} \tag{36}$$

and

$$y_{[a,b]}(z) = \begin{cases} 0, & z \le a, \\ L(z - a), & a < z < b, \\ L(b - a), & z \ge b. \end{cases} \tag{37}$$

Each inventory is continuous at both boundaries. The nonzero full-range tail adds L_0/z and $L_0 z$. Summing two nested ranges yields equation (9). It gives a continuous path even when active liquidity changes abruptly at a boundary.

Within a region, derivatives are

$$\frac{dU}{dz} = -\frac{L_A}{z^2}, \qquad \frac{dV}{dz} = L_A, \qquad -\frac{dV}{dU} = z^2 = p.$$

This establishes marginal price without assuming that price equals the raw inventory ratio. Inactive ranges retain one asset, and active ranges have offsets from their finite boundaries. Those offsets contribute to inventory value but do not alter the derivative ratio inside the region.

B.1 Boundary crossing

For base input, let the next lower square-root boundary be $a < z$. The amount to reach it is

$$\Delta U_{\text{boundary}} = L_A(1/a - 1/z).$$

If the remaining input is smaller, equation (10) finishes the quote. Otherwise the quote consumes this amount, credits the output $L_A(z - a)$, changes the active set, and continues. Quote input proceeds toward the next upper boundary with input $L_A(b - z)$. The full nested path contains only four boundaries, so at most five segments are traversed.

The Rust cache uses normalized integer quantities and precomputed branch geometry. Its construction and rounding need separate review from these real-valued expressions. In particular, the continuous equation is not an instruction to convert arbitrary raw token amounts directly to floating point in an SDK. Exact quotes should follow the program's normalization, direction, and conservative amount rounding.

B.2 A transparent illustration

The generator uses $L_0 = 100$, core liquidity 450 on $[0.95, 1/0.95]$, and shoulder liquidity 450 on $[0.85, 1/0.85]$. These are explicitly chosen square-root-price bounds. They are not raw governance width values and are not intended as a recommended market configuration. At $z = 1$, $U = V = 190$ and active liquidity is 1,000. Outside both ranges active liquidity is 100.

Consequently, center inventory can be much smaller than active curve liquidity, while the tail still holds inventory beyond the concentrated region. This is the geometric source of concentration. It is distinct from adding synthetic hLP claims to a trader's quote. The example plots inventory ratio V/U beside marginal price z^2 to make their separation observable away from the center.

The implementation's peak-amplification control measures center depth relative to a constant-product pool at corresponding center reserves. It should not be read as an arbitrary multiplier of every trade's output, every price region's depth, or every collateral value. Larger trades can cross into lower-depth shoulders and tails, and lending admission uses the tail-based conservative risk construction.

B.3 Controller economics

A change in the concentration center changes the geometry supported by a given inventory. The controller therefore needs both a target and an admission budget. Dusk retains eligible dynamic surcharge in a protected reserve bucket and evaluates a deferred target against the state in which it will be committed. Deposits, withdrawals, losses, and prior trades can invalidate an old cost estimate.

The ordering matters economically. First commit a previously admissible target, then price a user action from the resulting frozen state, and only then use the successful action to update future observations. This prevents the current action from unconditionally choosing a new center based on its own final price. It does not eliminate manipulation through a sequence of transactions or establish that every admitted movement is profitable under an external benchmark.

C Algebraic hLP Ownership

Let base hLP own the fraction E_b/U of the ordinary portfolio, measured before converting that fraction into integer shares. It therefore owns base claim E_b and quote claim E_bV/U. Matching the quote claim with quote funding debt leaves base equity E_b. Quote hLP has the symmetric construction. Adding ordinary and vault-owned inventories gives equation (21).

Factoring the endpoint reveals a useful identity:

$$X_b = U\left(1 + \frac{E_b}{U} + \frac{E_q}{V}\right), \tag{38}$$

$$X_q = V\left(1 + \frac{E_b}{U} + \frac{E_q}{V}\right). \tag{39}$$

Hence $X_q/X_b = V/U$ in exact arithmetic. This is an inventory-ratio identity. It does not equate that ratio with the concentrated marginal price. The inventory ratio is sufficient to match opposite claims; curve geometry remains necessary to price trades and express marked gross leverage.

C.1 A two-vault example

Take ordinary inventory $(U, V) = (1{,}000, 1{,}000)$, base equity $E_b = 100$, quote equity $E_q = 50$, and ordinary supply $S_o = 1{,}000$. Then

$$s_b = 100, \qquad s_q = 50, \qquad S = 1{,}150,$$

and total executable reserves are $(1{,}150, 1{,}150)$. Base hLP's quote claim and funding target are both 100; quote hLP's base claim and funding target are both 50.

Now apply a fee-free 100-base input to the ordinary constant-product tranche, while holding target equities fixed for this illustration. Ordinary output is $90.90909\ldots$ quote and the

endpoint is

$$U' = 1{,}100, \qquad V' = 909.09091\ldots.$$

The new canonical funding targets become

$$D_q^{h,*} = 100\frac{909.09091\ldots}{1{,}100} = 82.64463\ldots,$$

$$D_b^{h,*} = 50\frac{1{,}100}{909.09091\ldots} = 60.5.$$

Total executable reserves are approximately $(1{,}260.5, 1{,}041.73554)$. The trader did not receive output from a 1,150-by-1,150 constant-product pool: quoting the full synthetic total would have given a different result. Reconstruction changes the gross ownership and funding decomposition around the already executed ordinary path.

In integer shares, equation (22) floors the vault balances against the unchanged ordinary supply. This can change total yLP supply even though ordinary holders neither mint nor burn during the spot trade. Fee eligibility is frozen at the appropriate earlier state, and actual debt shares are converted through the debt index. A complete transition therefore carries more information than just the real-valued reserve endpoints shown above.

C.2 What the target does and does not hedge

For a base-target vault at a marked price p, gross collateral value in quote units is $E_b p + E_b V/U$. The second term is canceled by opposite debt in the canonical construction. Gross leverage then follows equation (23). At a constant-product point, $V/U = p$ and the ratio is two. At a concentrated point, V/U can be above or below p, and the two target vaults have different gross leverages.

The familiar constant-product LP value proportional to $\sqrt{p}$ provides motivation for continuously adjusted leveraged liquidity. A static two-times LP position exposed to a discrete jump is not the same process as a continuously adjusted position, and neither expression is the current Dusk execution algorithm. The earlier formula $E_0(\sqrt{r} - 1)^2$ should not be used to infer an active Dusk pre-solve threshold or quote correction. The present program quotes ordinary liquidity and reconstructs ownership algebraically.

Funding interest and fees can change the target equities used in reconstruction. Recovery explicitly decreases one target equity. A loss socialized by another path can also affect the economics of the vault's yLP claim. Opposite-asset neutrality at one endpoint therefore does not imply a constant token balance across time or an unconditional impermanent-loss guarantee.

C.3 Inventory weights and local leverage

The leverage formula also follows from the local sensitivity of an LP portfolio. At fixed geometry and before fees, let its quote value be $W = pU + V$. Since the marginal curve

satisfies $p\,dU + dV = 0$,

$$dW = U\,dp, \qquad \frac{dW}{W} = w_b \frac{dp}{p}, \qquad w_b = \frac{pU}{pU + V}. \tag{40}$$

The base inventory weight w_b is the LP's local percentage sensitivity to a base-price move. An opposite-funded position with gross leverage $1/w_b$ has unit local sensitivity before funding costs. Its leverage is therefore $1 + V/(pU)$, matching equation (23). In base-denominated value, the symmetric quote weight is $w_q = V/(pU + V)$ and the corresponding leverage is $1/w_q = 1 + pU/V$.

At a constant-product point $pU = V$, both weights are one half and both leverages equal two. Concentration changes inventory weights away from that balance, even though the same marginal relation $p\,dU + dV = 0$ holds. This derives the current inventory-dependent leverage from the same LP-payoff intuition used in Section 5. Local sensitivity and the finite-move tracking gap are different statements; neither removes the cash and funding constraints on actually reconstructing the position.

C.4 Settlement preconditions

The integrated engine verifies the expected curve revision, yLP supply, each vault's yLP shares, indexed debt shares, and principal state. It settles the specified funding-interest cash debit, replaces the old hLP live component with the endpoint's funding component, removes current public and isolated unpaid interest for the endpoint comparison, and verifies the bounded raw-atom distance. It then commits ownership and debt and asserts the reserve ledger.

The current integrated preparation also requires hLP backing inventories to be empty; direct exits account for backing inventory when present. Positive marked NAV alone does not establish that a particular operation is admissible. Available cash, funding-interest settlement, the prepared pre-state, and the instruction's other guards still apply. If these constraints cannot be satisfied, the transaction can fail atomically rather than publishing a partial hedge.

D Recovery Incentives and the Terminal Boundary

Recovery distinguishes three quantities: actual indexed debt D, canonical opposite claim A, and legally usable revenue Y. The net gap is $g = [D - Y - A]_+$. The critical flag instead compares actual D with $9A/8$ before subtracting revenue. These tests answer different questions: the net gap determines the incentive currently needed, while the actual ratio identifies the debt stress boundary.

For $A = 1{,}000{,}000$ raw units, the following examples have no usable revenue and a one-to-one target conversion. Assume matching input is at least the gap and target equity is large enough to pay the bonus.

Debt	Net gap	Discount (bps)	Bonus (atoms)	Critical
1,002,499	2,499	0	0	No
1,002,500	2,500	20	5	No
1,062,500	62,500	500	3,289	No
1,125,000	125,000	500	6,578	Yes

At a five-percent discount, an amount with fair target value v buys total target value $v/0.95$, so the incremental bonus is $v(0.05/0.95)$. Multiplying by 0.05 alone understates the bonus. The implementation floors the conversion and then the discount arithmetic, caps matched input at the gap, and caps the bonus at available target equity. These caps do not remove the ordinary curve impact or fees on the complete input.

D.1 Revenue can change the incentive without changing the critical flag

For example, let $D = 1{,}130{,}000$, $A = 1{,}000{,}000$, and usable revenue $Y = 120{,}000$. Actual debt exceeds $9A/8$, so the critical flag is true. The net gap is only 10,000, giving an 80-basis-point discount before the other caps. If usable revenue covers the entire excess, the net gap is zero and no positive recovery bonus is produced, even though the actual-debt critical test remains true. The flag is diagnostic; an ordinary swap can earn a bonus whenever the net gap and matching input support one.

When the claim is zero, actual debt can still imply critical stress, but ordinary recovery pricing returns no positive incentive from the ratio formula. This avoids dividing by zero or using an undefined conversion to fund an output. Terminal closeout and the broader state checks, rather than an extrapolation of equation (26), determine the admissible next action.

D.2 Terminal closeout is an interest-loss path

The terminal engine checks that marked collateral is below indexed funding debt and that the shortfall is no larger than the unpaid interest. This distinguishes passive funding insolvency from a principal hole. The plan is bound to the state it valued, records the permitted insurance request, retires the vault's owned yLP, and realizes the recoverable interest. Insurance is senior to the remaining socialized interest loss and is constrained by available capital, protocol concentration limits, and caller bounds.

The caller also supplies a maximum socialized-loss bound and minimum bounty output. The bounty is computed from recovered funding interest; it is not an extra source of recovery capital. Published user yield remains a separate liability. A surviving hLP token balance can be burned for zero principal after a terminal event, while its checkpointed yield can still be harvested.

This mechanism provides an explicit action once funding growth has exhausted equity, but depends on someone submitting an admissible transaction. A bonus or bounty does not prove that an economically motivated caller will exist for every token, especially when transfer costs,

fees, congestion, or small amounts dominate. These are liveness questions for stress testing and operational observation.

E Admission, Eligibility, and Settlement

Credit admission, liquidation eligibility, and liquidation pricing deliberately use different functions. Keeping their roles separate prevents an appealing but incorrect statement that "all risk uses the same AMM output formula."

Decision	Valuation role
New public credit	Conservative tail-based shadow CPMM, lower directional/symmetric collateral price, pessimistic depth, position and aggregate underwriting checks.
Liquidation eligibility	Linear collateral mark at symmetric EMA compared with indexed debt and the position's stored liquidation collateral factor.
Ordinary flash purchase	Bound net debt payment plus charge for an EMA-valued collateral slice at an affordable discount.
Emergency internal exchange	Concentrated reference curve at symmetric EMA and smoothed depth; real inventory, cash, insurance and writeoff settle separately.

E.1 Admission and collateral factors

Equation (30) is concave in collateral size. Its derivative is

$$W'(c) = \frac{\widehat{x}\widehat{y}}{(\widehat{x}+c)^2}, \qquad W''(c) = -\frac{2\widehat{x}\widehat{y}}{(\widehat{x}+c)^3} < 0.$$

It recognizes less than the linear mark $c\widehat{y}/\widehat{x}$ for every positive collateral amount. That property alone does not prove that dividing collateral among accounts cannot expand total credit drawn: a concave function applied separately can produce a larger sum. The market-level underwriting contributions and global checks must also be considered. The paper therefore does not claim account-splitting resistance solely from the one-position formula.

A credit draw stores a newly computed liquidation collateral factor f. The user can specify a minimum acceptable value; credit capacity is buffered below the accepted factor. In fraction notation, the buffered factor is $0.95f$. On collateral removal, the implementation takes the greater of the stored factor and the newly calculated factor before checking remaining collateral. These are action-specific rules, not a continuously reset liquidation factor after every market movement.

Each public debt side uses the borrower's opposite-asset collateral. The aggregate health book recognizes a debt-bearing contribution capped by underwriting logic, rather than claiming that all idle collateral across accounts is locked for other borrowers. Removing a contribution from a global summary does not move another account's tokens. Exact debt-share rounding and contribution updates are important coupled-state checks in a source review.

E.2 Observation timing and the public credit budget

An internal EMA can be represented conceptually as $m' = \alpha m + (1 - \alpha)x$, with decay determined by elapsed slots and a half-life. Dusk has separate observations for price, directional price, curve depth, and concentration center. Their role is to avoid treating every instantaneous favorable movement as immediately recognized credit capacity. They cannot establish that the market has truthful external information.

The public credit limiter is a continuously draining usage bucket over a 24-hour window, not a UTC-midnight reset. Gross new public principal consumes its available budget. Debt repayment restores cash but does not immediately undo gross usage. The implementation carries fractional leakage so repeated updates do not lose all sub-unit drain. This property is specific to the limiter and should not be generalized to every fixed-point mechanism in the protocol.

E.3 A liquidation payment and loss example

A partial flash buyer repaying 100 debt tokens owes a separate one-token solvent charge. At a 0.5% applied discount and unit reference price, ignoring token fees, collateral debit is $101/0.995 = 101.5075377$. At the 5% ceiling it would be $101/0.95 = 106.3157895$. The larger debit can worsen the survivor's health; the program selects an affordable rate or rejects that partial. The ramp therefore quotes a maximum, not a promise that every slice receives five percent.

A committed eligible observation starts the incentive episode, with a two-minute ramp and no mandatory waiting period. An atomic flash begin cannot start a durable episode if that same transaction fails. Recovery and account recreation invalidate stale episode use. The emergency 70%-of-maintenance permission is independent of elapsed time, so a continuously unprofitable position above that boundary need not eventually become emergency-executable.

For a completed full settlement with 900 principal, 50 interest and 850 net proceeds after caller/transfer costs, the eligible principal gap is 50. Insurance at 75% supplies at most 37.5, subject to its cash and shared window budget; 12.5 principal is written off and 50 interest canceled. A shortfall does not earn the additional solvent liquidation charge. A 50% window cap limits insurance spending, not the market's possible loss. Existing hLP terminal eligibility and coverage remain a separate claimant policy within the same budget.

The internal reference exchange transfers collateral into pool inventory and offsets debt. Later price discovery can change that inventory's marked value. In the accepted #287652 analysis, a second decline in marked pool resources after trading does not represent another unresolved principal balance. Counting all changes in LP claims as cash paid to an arbitrageur would also misclassify transfers into protected recenter reserves. The retained closeout compares fees, acquisition cost, insurance, principal writeoff, LP/hLP claims and protected resources at a common external mark.

F Rate Controller and Update Cadence

The implemented defaults and accepted governance range are:

Parameter	Default	Range or bound
Target utilization	70%	60%–75%
Curve steepness κ	4	2–8
Anchor adjustment speed	20/year	1–50/year
Initial anchor APR	4%	Initialization value
Anchor APR	Adaptive	0.1%–200%
Anchor multiplicative step	State-dependent	Factor 0.5–1.5 per update
Accrued elapsed time	Slots $\times$400 ms	At most one year per update

With the default steepness, a four-percent anchor implies one-percent APR at zero utilization and sixteen-percent APR at full utilization. At the maximum 200-percent anchor, full utilization gives 800-percent instantaneous APR under the default steepness. The anchor bound cannot be advertised as the user's maximum debt rate. Governance changes to steepness change this multiplier range.

The implementation first computes utilization and the instantaneous rate using debt, cash, and the current anchor. The index increment has the form

$$I' = I + \left\lfloor \frac{I \lfloor r_{\mathrm{NAD}} \Delta t / T \rfloor}{\mathrm{NAD}} \right\rfloor,$$

where T is milliseconds per year and Δt is capped elapsed time. It then adapts the anchor with equation (34). Fixed/public and isolated indexed debt are converted separately; summing shares before a single conversion would change some raw-atom boundaries. hLP funding debt uses the same asset-side index but has different live-reserve treatment.

The daily illustration uses $\tau = 1/365$ for thirty updates and holds utilization fixed by assumption. It records the APR after each anchor update, not a borrower's realized annual return. Fixed utilization would itself require compensating supply or debt changes as interest accrues; the example isolates the controller rather than simulating those agents.

For comparison, collapsing thirty days into a single update at full utilization gives $r'_T = 0.04(1 + 0.5) = 0.06$ because the step is clamped. Thirty daily updates produce a different anchor. Even below the clamp, repeated linear factors and a single larger linear factor are not identical. This makes update cadence a disclosed experiment parameter. It also motivates exact-program testing of very frequent and very sparse activity rather than inferring frequency independence from the controller's intended exponential intuition.

With no debt shares, the program moves the anchor toward its lower bound and advances the accrual clock, without generating debt interest. When hLP debt exists, funding APR observations include its current asset-side rate and update the vault's twelve-hour funding-rate EMA. That signal is also used by hLP conditional stop-rate orders, tying the order threshold to a smoothed funding measure rather than the instantaneous anchor alone.

G Settlement, Delegation, and Economic Boundaries

G.1 Fee basis and token transfers

Let a be the amount credited to reserve custody after an input token transfer. For input-denominated fees, the total Dusk fee budget is at most $\lfloor a/2 \rfloor$; executable input is reduced by the priced fee. For an output-denominated fee, the prepared curve produces gross output b and the fee budget is at most $\lfloor b/2 \rfloor$. User output is reduced by the withheld fee and then by any token-program output transfer effect.

These definitions matter when quoting a minimum received amount. A bound on protocol fee share does not bound AMM price impact. It also does not erase a Token-2022 transfer fee. A margin entry limit uses the all-in relation between paid notional and collateral actually obtained, with direction-appropriate rounding. A client that treats gross output, fee-net output, and final token credit as the same amount can accept a worse trade than the user intended.

Fees also need correct ownership timing. The trade's eligible ordinary supply and vault-owned yLP balances are captured before later reconstruction changes those balances. Claimable yield is published through its liability indexes. Compounded yield increases principal and changes the target equities through the integrated fee allocation. The integrated hLP full-value path marks opposite-token fee conversion at the actual concentrated marginal price. The retained native replay below used the earlier endpoint-inventory-ratio conversion; its fee-allocation numbers are historical and must not be substituted for current quotes. Protected recenter funds remain a separate custody claim. These are three different destinations for collected value, each with a different effect on future quotes and redemptions.

G.2 Margin and delegated orders

A public credit draw leaves the principal curve coordinate unchanged at its cash/debt step. Isolated margin composes that step with a real spot trade. This makes entry immediately sensitive to depth, fees, and cash availability. The resulting collateral and debt belong to a dedicated position, and its closeout value is recomputed from executable market machinery. Risk increases must satisfy stored and newly retained initial-equity obligations. Standalone margin withdrawal and the fixed two-percent unwind rejection are removed. Debt repayment retains the held asset; proportional owner closure returns net residual. Section 6.5 specifies the current action model.

The delegate program adds conditional authority around Dusk instructions. A take-profit or stop-loss condition, a margin-entry price limit, or an hLP funding-rate threshold narrows when an executor can act. The underlying market instruction still validates its accounts and settlement. Conditional permission is not reserved liquidity, guaranteed execution, or a promise that a threshold crossing will be observed in time.

hLP orders escrow the hLP shares while preserving the original owner's yield recipient. On execution, the output bounty is charged before evaluating the configured owner payout floor; final transfer fees remain relevant. After cancellation or execution, the dedicated yield-settlement

path can harvest remaining liabilities and close the appropriate order state. The escrow's token ownership should not be interpreted as transferring economic entitlement to already assigned user yield.

G.3 Owner, recipient, and harvest authority

The optional harvest authority is stored separately from the yield recipient. The owner can grant or revoke it without changing where funds go, and can change the recipient without implicitly changing the independent authority. Harvest authorization accepts the owner, recipient, or configured authority. Account validation restricts payout to the recipient's canonical associated token account.

This division is useful for program-owned positions and automated collection. It is also a source of documentation error if described simply as "permissionless harvest" or "only the owner may harvest." The current rule is an explicit set of authorized signers with a constrained destination. A program-derived owner still needs its program to sign the invocation for owner-only configuration changes.

H Native Trade, hLP Reconstruction, and Who Bears the Difference

The companion `simulations/native-hlp-flow.json` retains eight executions of the earlier #45 source snapshot `5c0aca7f` using the native benchmark API. It tests amplification one and four, matched runs with/without both hLPs, and zero/30-basis-point base fees with claimable/compounded LP allocations. Tokens have nine decimals; the trades use the same slot, so funding accrual is zero. Dynamic surcharge and token transfer fees are zero in this isolated comparison. These values illustrate the inventory and holder-value identities; they are not current integrated-branch fee quotes, token-CPI tests or validator execution. The combined-branch instruction tests are separate evidence.

H.1 The illustrative two-dollar shortfall

The toy concentrated segment $(x + 100)(y + 100) = 40{,}000$ starts at $x = y = 100$, $p = 1$. Selling 22.222222 META receives 20 USDC and ends at $x = 122.222222$, $y = 80$, $p = 0.81$. The LP is worth 179 instead of the 181 from holding its original assets. The trader received 20 for input worth 18 at that common final mark. The difference is two dollars before fees. Subtracting unchanged debt 100 gives static funded equity 79 versus a 100-META holding worth 81. This is the toy concentrated segment; the earlier pure-CPMM static example gives a one-dollar gap at the same price ratio. Neither is an extra two-dollar invoice paid to Dusk hLP.

H.2 An executed concentrated example

Ordinary yLP contributes 100 META and 100 USDC. Each hLP contributes 50 target tokens. With META initially one USDC, actual cash is 150 META plus 150 USDC; gross live claims are 200 of each, including 50 of each internal funding component. The concentration setting is peak amplification four, core half-width 100 bps and fade width 400 bps. A trader sells 22.222222222 META, receiving 19.360948321 USDC without fees. Final price is 0.711853017 and actual cash is 172.222222222 META plus 130.639051679 USDC.

Holder	Gross claim and matched debt after settlement	Net USDC value
Ordinary yLP	122.222222 META + 80.639052 USDC; no debt	167.643309
META hLP	50 META + 32.988703 USDC; owes 32.988703 USDC	35.592651
USDC hLP	75.783519 META + 50 USDC; owes 75.783519 META	50.000000

The META vault's USDC claim and debt fall together; the USDC vault's META claim and debt rise together. These are funded ownership changes, not additional external payments. Gross leverages become approximately 1.926840 and 2.078935, while net target backing remains 50 tokens each. The same price move at fixed concentration settings therefore changes the represented leverage.

Holding the ordinary depositor's original assets would be worth 171.185302. Its 3.541992 shortfall equals the trader's marked gain $19.360948321 - 22.222222222(0.711853017)$. Both hLPs match holding their respective 50 target tokens within rounding. Across all holders, net claims sum to actual cash value. Matched zero-fee runs without hLP give the identical ordinary inventory and trader output; the replay does not identify an additional hLP subsidy on that trade. Calling the marked trader gain arbitrage profit additionally assumes an external venue at the common final mark and excludes external acquisition/execution costs.

H.3 Fees and the boundary of this result

With a 30-bps base fee and 20% protocol share, the same gross input pays approximately 0.066667 META. Before ownership reconstruction, the initial ordinary/META-hLP/USDC-hLP shares split the LP allocation 50/25/25. With compounding disabled, ordinary yLP receives a 0.026667-META entitlement, each hLP 0.013333 META, and protocol 0.013333 META. Final holder values including these pro-rata claims are approximately 167.764783, 35.644084 and 50.009503 USDC at the new final mark 0.712691622.

With LP compounding enabled in this historical replay, the former reserve-ratio conversion credits USDC hLP 0.008807 USDC for its opposite-token fee, compared with approximately 0.009503 at the frozen marginal price; the difference stays in ordinary inventory. The integrated branch uses the concentrated marginal mark instead. Funding accrual, public receivables, recovery rewards, market recentering and debt losses are separate transitions and cannot be inferred to be costless from the zero-accrual examples. The current accounting and validation boundary is described in Section 7.

I Implementation Evidence and Reproducibility

The runtime integration candidate combines PR #45 at `04941f19` and PR #52 at `2f46e0e7`. The retained numerical replay uses earlier #45 source at `5c0aca7f`; its identifier is recorded in `paper/SOURCES.md` and generated data metadata. The table below identifies semantic entry points in the integrated candidate; it is not a substitute for the surrounding account checks and caller ordering.

Claim	Implementation anchor (under `programs/dusk/src/`)
Stored reserve identity	`transitions/ledger.rs`: `assert_virtual_reserve_invariant`.
Unpaid-interest exclusion	`transitions/amm/mod.rs`: `unrealized_interest`, `curve_reserve`.
Nested curve	`transitions/amm/curve.rs`: geometry preparation and analytic segment quotes.
hLP ownership	`transitions/liquidity/hlp/integrated.rs`: `reconstruct_hlp_endpoint`, `reconstruct_hlp_ownership`.
hLP settlement	`transitions/liquidity/hlp/engine.rs`: identity-bound transition, cash floors, terminal waterfall.
Recovery discount	`transitions/liquidity/hlp/recovery.rs`: `quote_hlp_recovery`.
Rates and debt accrual	`math/risk.rs`; `transitions/lending/mod.rs`: `accrue_side`.
Flash/emergency	`instructions/flash_liquidation.rs`; `instructions/emergency_liquidation.rs`; `transitions/liquidation_policy.rs`; `transitions/amm/liquidation.rs`.
Isolated margin	`transitions/leverage.rs`; corresponding instruction handlers.
Custody and harvest	`instructions/accounts.rs`; `instructions/liquidity/harvest.rs` and `state/yield_account.rs`.

Some functions and fields retain historical names. A helper called "live," "rebalance," or "tracking" is not evidence that the earlier paper's algorithm remains in use. The review followed the active callers and distinguished test-only helpers from executable paths. The old tracking-loss solver and generic test-only fee routines should not be used to describe current integrated spot trades.

I.1 Rebuilding the illustrations

Run the generator from the repository root with a Python environment containing ReportLab. Its arithmetic uses the Python standard library; ReportLab is used only to create vector charts. The script writes four CSV datasets, machine-readable assumptions, three LaTeX tables, and three PDF figures. It checks range-boundary continuity, depth scaling, the regional invariant, hLP ownership, rate anchor bounds, recovery thresholds, and selected accounting examples before writing its validation receipt.

Compile either top-level LaTeX source from the `paper/` directory using Tectonic. Both files include the same `main.tex` and `preamble.tex`; the expanded version enables additional discus-

sion and includes these appendices. This organization prevents a parameter correction in one paper from leaving the other with an incompatible equation. The bibliography contains only sources actually used in this revision.

The numerical checks establish properties of the disclosed illustrations. They are not differential tests between Python and every Rust instruction. The source review and the repository's Rust and instruction tests provide separate implementation evidence; the historical CI receipt belongs to its original revision. This integrated revision changes the program interface by removing the dedicated rescue instruction.

I.2 What a stronger evaluation would measure

A useful replay should run the exact current program through multiple boundary crossings, both fee denominations, asymmetric token decimals, funding accrual, hLP recovery, flash settlement, and EMA-priced emergency execution. Report cash by side, debt by domain, ordinary reserve depth, total reserve claims, hLP target equity, interest paid, insurance draw, and socialized loss. Net economic returns require trading fees, funding cost, execution costs, and the chosen external benchmark, not just a reserve-identity check.

Adversarial sequences should include favorable price manipulation before drawing credit, temporary depth provision, collateral withdrawal after a stored-factor update, many small accrual intervals, scarce cash during hLP interest settlement, and concurrent fee compounding with a large spot trade. A review should distinguish a rejected transaction from a successful but loss-making one. Rejection can preserve accounting while reducing market availability; a successful transaction can preserve every identity while transferring economic value between participants.

For flash liquidation and recovery, participation must be modeled explicitly. An attractive theoretical price does not ensure a filler has the required asset or can profit after execution costs. Measure time to an admissible fill, remaining collateral, realized output, available insurance, and the residual loss. For hLP terminal closeout, separately verify the full marked-value and EMA insolvency preconditions, principal-first loss waterfall and preservation of funded holder yield.

No claim of current comparative performance is imported from the earlier constant-product or prototype concentrated-curve studies. Those experiments evaluated different mechanisms. A new comparison would need a common workload, precisely matched capital and risk assumptions, current executable artifacts, and a disclosed source revision for every system. Until then, the external references in this paper provide mathematical and architectural context rather than a performance ranking.

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